

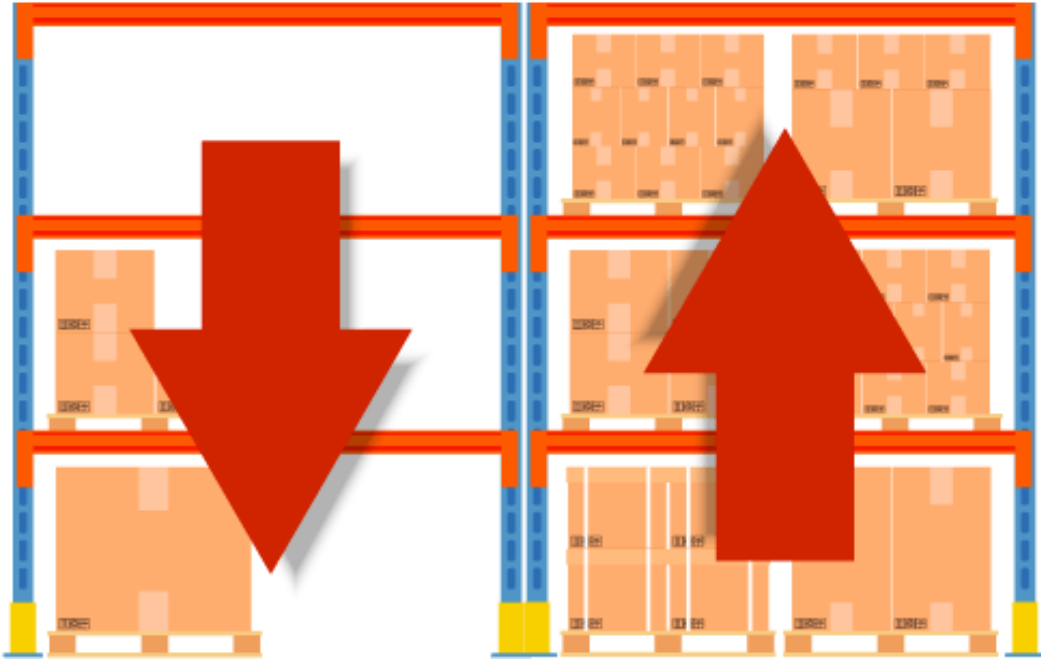


PACCAR INVENTORY OPTIMIZATION

*DATA-DRIVEN SOLUTION TO OPTIMIZE
INVENTORY AT PACCAR PARTS*

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THE PROBLEMs:

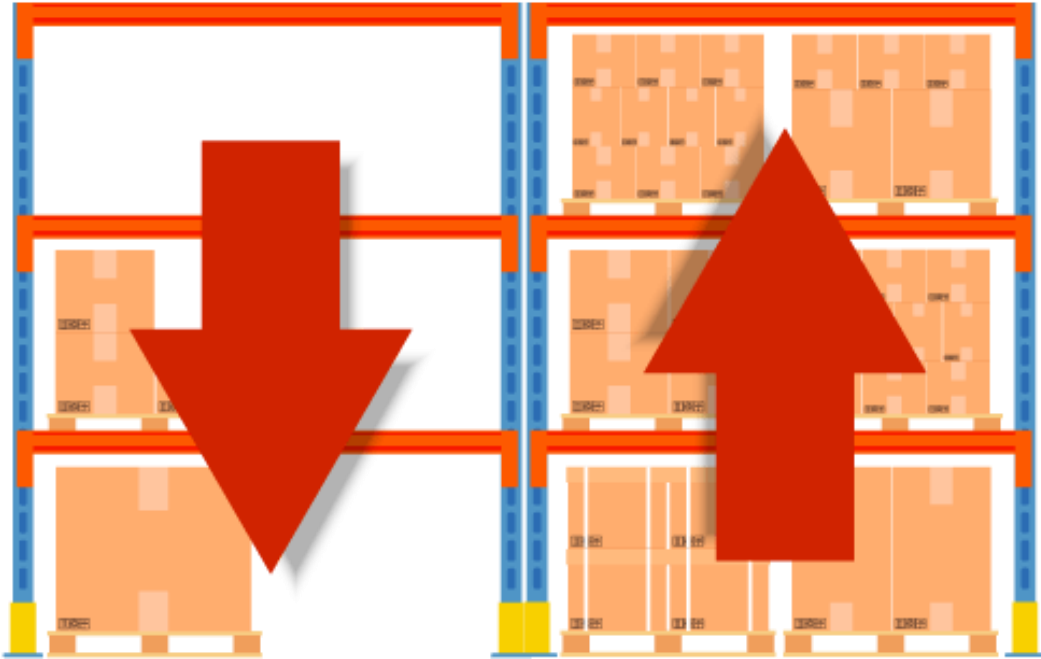


Inventory Challenges
(Supplier Reliability, Lead
Time, On-transit item)



Manual Review

THE PROBLEMs:



Inventory Challenges
(Supplier Reliability, Lead
Time, On-transit item)

Parts Above Safety Stock Still Stocked Out — Due to Longer Delivery Times



“

This shows the opportunity to enhance the model by layering in lead time, especially for items with slower supply recovery

”

In-transit Inventory Contributes to Stockouts Despite Availability



“ The gap between inventory on-hand and inventory on-the-way contributed to unexpected stockouts”

Late Supplier Deliveries Are a Hidden Cause of Regional Stockouts



Marked Safe (Above Safety Stock)



SUPPLIERS 15% delay on average



Marked Risky (Below Safety Stock)



SUPPLIERS 18% delay on average

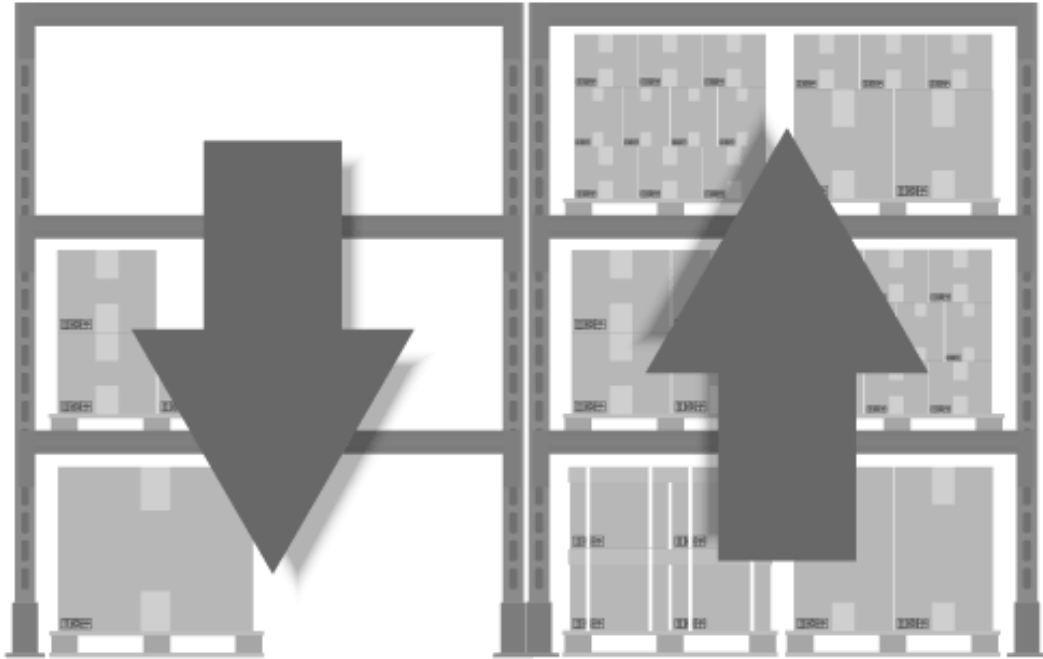


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The system sees available inventory, but not whether incoming supply will arrive on time — and that’s where risk can slip through

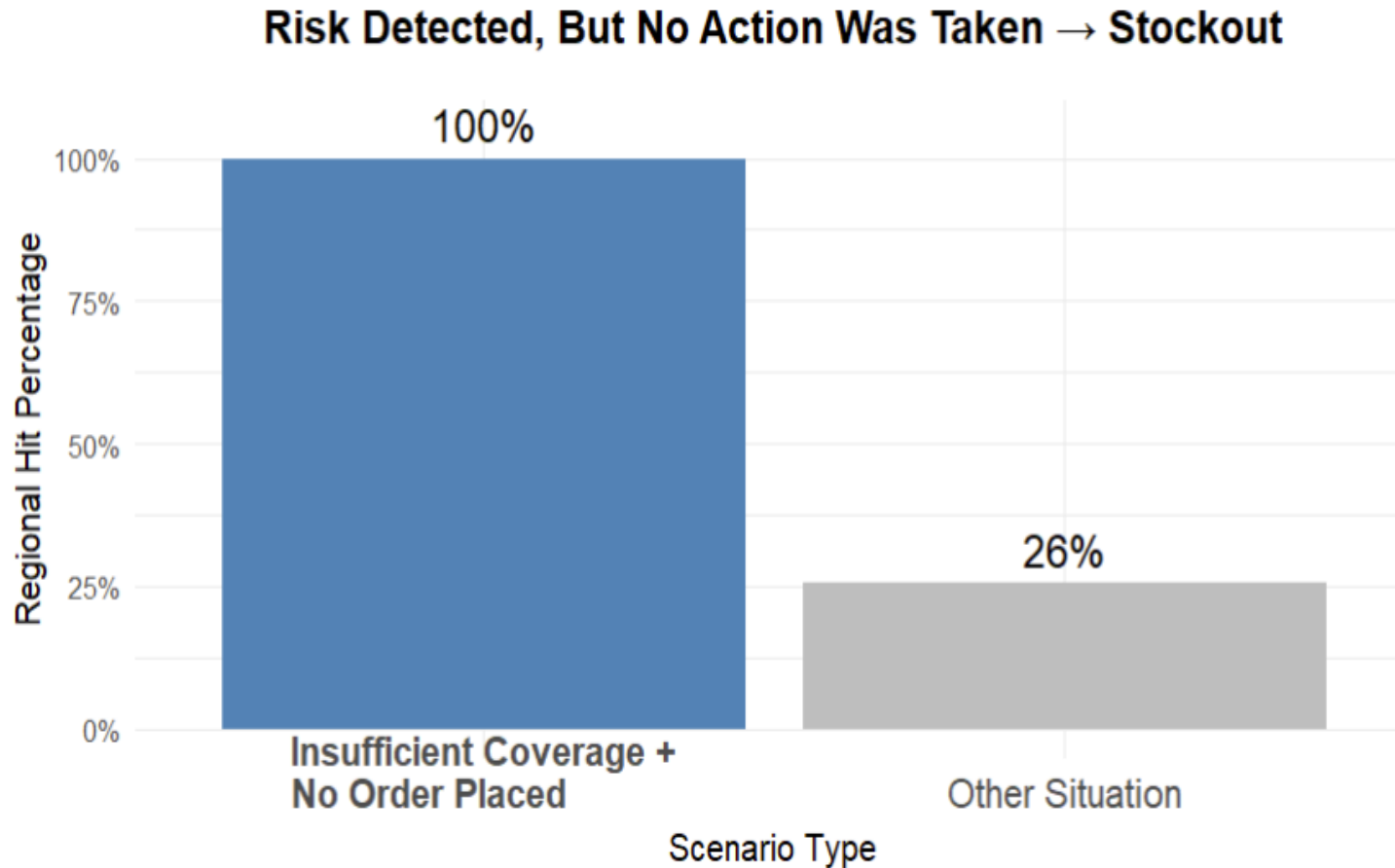
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THE PROBLEMs:



Manual Review

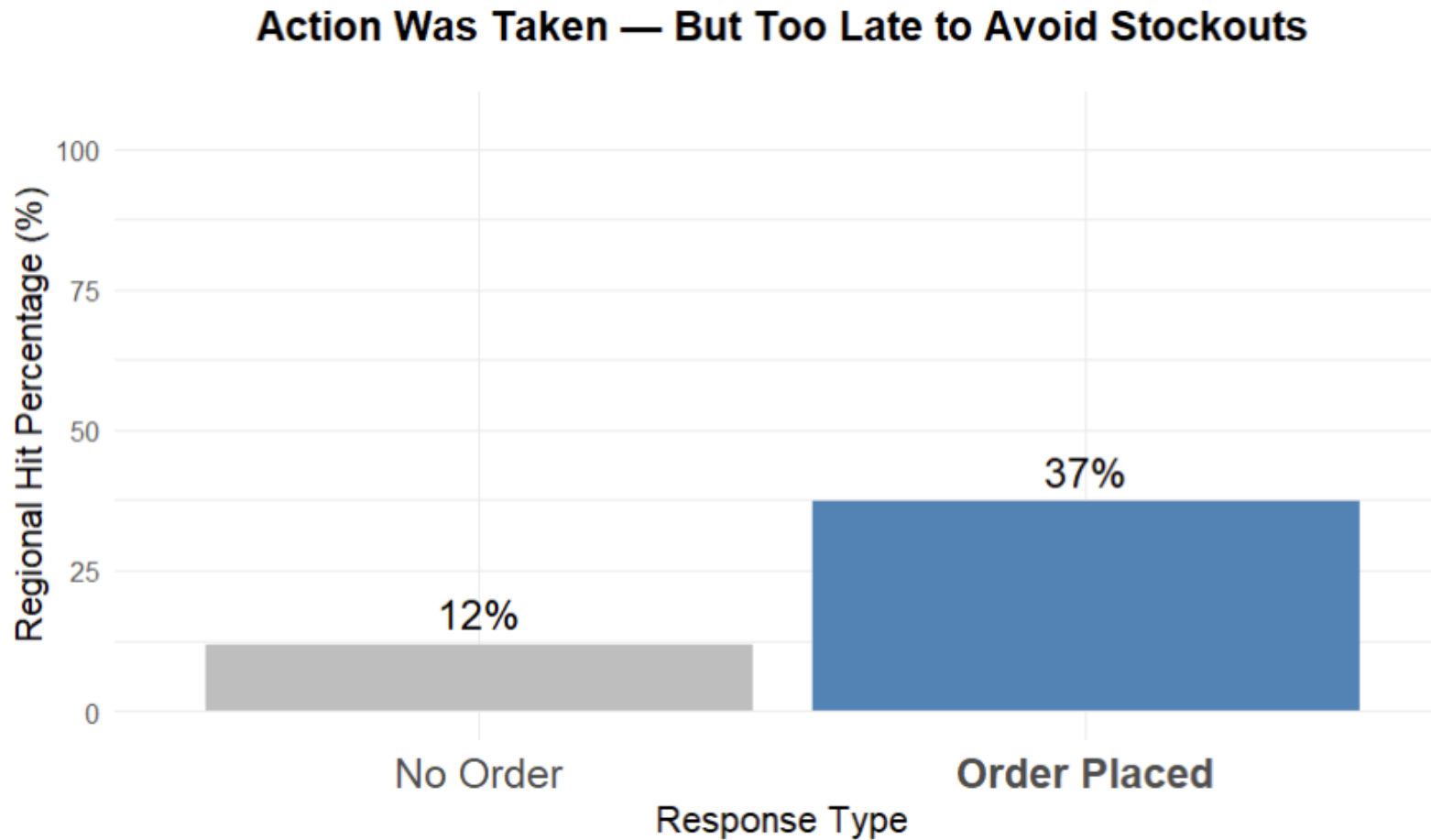
When Manual Review Didn't Act Fast Enough...



“If inventory coverage is low and no order is placed, stockouts occurs 100% of the time.”

*** Insufficient coverage = stock on hand will not last until the next shipment arrives (days on hand to lead time < 1)*

When Manual Review Didn't Act Fast Enough...



“Even when an order is placed, it’s often too late to prevent a stockout. Earlier detection helps planners respond in time.”

Our Solution: Predictive Model for Proactive Ordering

Model Objective

- Predicts whether a RHIT will occur (rhit_label = 1)
- Uses Gradient Boosting with inventory , supplier, and demand signals
 - Trigger action when:
Low coverage (e.g. < 14 days) + high RHIT probability

Action Threshold Table

Coverage Days	RHIT Probability	Action
< 14 days	> 30%	 Order TODAY
< 21 days	> 40%	 Order This Week
< 30 days	> 50%	 Monitor Closely
≥ 30 days	—	 Normal Monitoring

Example:

Part: A

RHIT Probability = 55%

Coverage = 12 days

Supplier reliability = Low

→ Triggered:  Order TODAY

Key Results – Early Warning Performance

Baseline

67%

Accuracy

Accuracy of “High Risk”
and “Low Risk” Predictions

51%

Precision

Accuracy of “High Risk”
Predictions

Improvement Model

80%

▲ 13%

Accuracy

Accuracy of “High Risk”
and “Low Risk” Predictions

66%

▲ 15%

Precision

Accuracy of “High Risk”
Predictions

70%

Recall

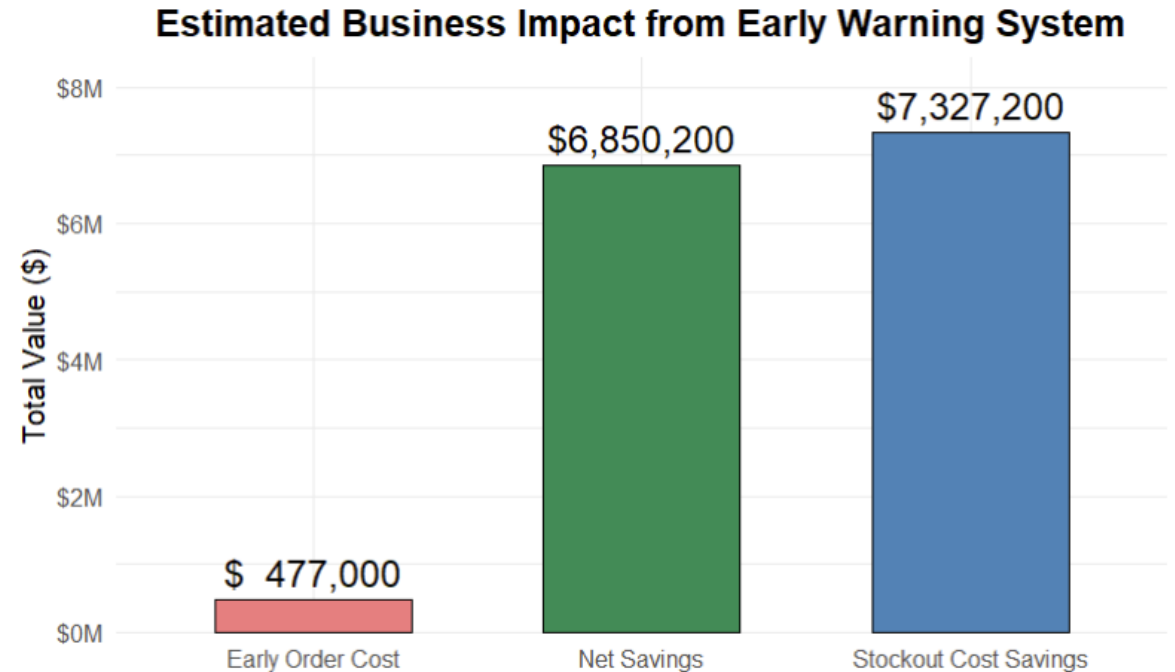
Business Value of Model with Early Warning System

✓ What Our Model Prevented

Metric	Value
RHITs Caught Early (Recall)	6,106 items
Est. Cost per RHIT	\$1,200 (emergency freight, downtime, admin)*
→ Stockout Cost Savings	\$7.33 Million*

⚠ What It Cost to Act Early

Metric	Value
False Alarms (no RHIT, but flagged)	3,180 items
Est. Cost per Early Order	\$150 (storage, shipping inefficiency, admin)*
→ Early Order Cost	\$477,000*



Net saving = Stockout Cost Savings - Early Order Cost

*Cost values are estimates based on freight shipping and related expenses.

Recommendations for Future Development



Cross-Part Relationships

Demand for part B rises when part A is bought.
Demand shifts to part Y if part X is out of stock.



Real-Time Adaptation

Link live inventory, lead times, and backorders.
Incorporate external data (delays, tariffs, strikes).



Cost-Aware Prioritization

Adjust alerts based on freight cost, part value, and urgency.
Avoid over-prioritizing low-value items.



Deep Learning for Smarter Forecasting

Use deep learning to identify demand and stock patterns over time, focusing on fast-moving parts.

Thank you for listening

Appendix

The errors in our model

- **Oversimplified Time Handling:**

- The model uses a fixed 14-day window ($\text{coverage_days} < 14$) to trigger urgent actions, which doesn't account for varying lead times. This can lead to late orders for items with long lead times.
- This deviates from the goal of *preventing* stockouts, as it may not provide enough time to replenish inventory before it runs out.

- **Lack of Demand Uncertainty Handling:**

- The model uses coverage_days , which is based on forecasted demand, without fully accounting for forecast inaccuracy or variability.
- This can lead to unnecessary early orders (if forecasts overestimate demand) or stockouts (if forecasts underestimate demand).
- This hinders the goal of *cost-effective* inventory management.

- **Oversimplified Risk Assessment:**

- The risk_multiplier uses fixed thresholds for risk factors like supplier past-due percentage, which might not accurately capture the nuances of risk.
- This can lead to inaccurate prioritization of actions.

Model Code Conditions:

- **Coverage days (coverage_days) thresholds:**

Urgent action if coverage < 14 days

Soon action if coverage < 21 days

Monitor if coverage < 30 days

- **Lead time urgency flag:**

must_order_in_2wks = 1 if (days_on_hand - lead_time) < 14

- **Risk multipliers:**

Increased risk if supplier_past_due_pct > 0.3 (multiplier × 1.5)

Increased risk if velocity == '1' (multiplier × 1.2)

Increased risk if network_avail < 0.5 (multiplier × 1.3)

- **Priority levels based on adjusted probability (probability × risk multiplier) and coverage days:**

- RED: coverage < 14 and adjusted_prob > 0.3
- ORANGE: coverage < 21 and adjusted_prob > 0.4
- YELLOW: coverage < 30 and adjusted_prob > 0.5
- GREEN: normal monitoring otherwise

Model Code Conditions (table)

Condition Type	Details
Coverage thresholds	<14, <21, <30 days for urgent, soon, monitor
Lead time urgency	(days_on_hand - lead_time) < 14
Supplier risk	supplier_past_due_pct > 0.3 multiplies risk by 1.5
Velocity risk	velocity == '1' multiplies risk by 1.2
Network risk	network_avail < 0.5 multiplies risk by 1.3
Priority levels	Based on adjusted probability and coverage days